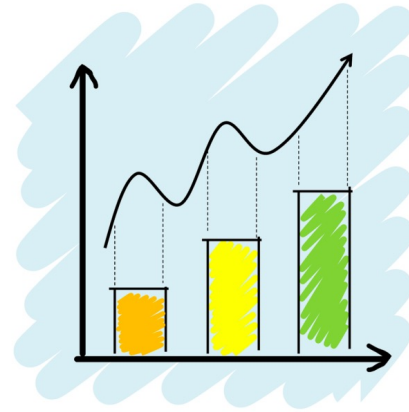




## Statistics

| $x$ | $y$ |
|-----|-----|
| 1   | 1   |
| 2   | 4   |
| 3   | 9   |
| 4   | 16  |

$$y = x^2$$

model  
formula  
function

```
def calculate_y(x):
```

```
    return x ** 2
```

distribution ←

```
y1 = calculate_y(5)
```

```
y2 = calculate_y(6)
```

```
y3 = calculate_y(7)
```

```
...
```

Categorical

↳ data containing categories

① City → S S K P N M..

② own-car? → Y N

③ Gender

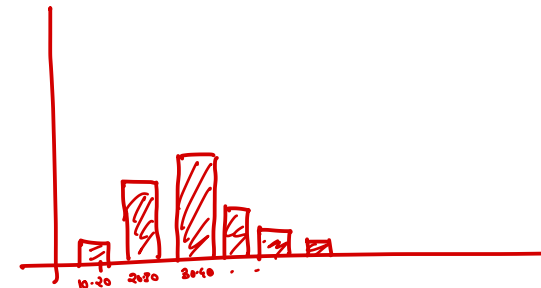
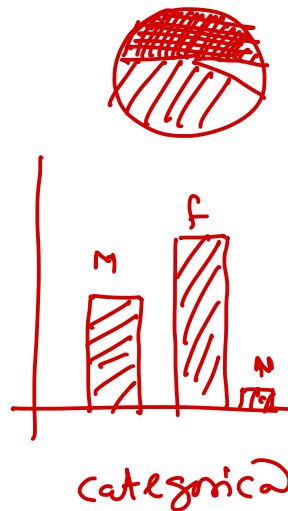
④ Education

⑤ O.S.

⑥ blood group

# Distribution

- The distribution of a statistical data set (or a population) is a listing or function showing all the possible values (or intervals) of the data and how often they occur
- When a distribution of categorical data is organized, you see the number or percentage of individuals in each group
- When a distribution of numerical data is organized
  - they're often ordered from smallest to largest, broken into reasonably sized groups (if appropriate)
  - then put into graphs and charts to examine the shape, center, and amount of variability in the data
- Types
  - Discrete probability distribution
  - Continuous probability distribution



# Terminologies

- **Frequency distribution**

- A frequency distribution is a table that displays the frequency of various outcomes in a sample.

- **Relative frequency distribution:**

- A frequency distribution where each value has been divided (normalized) by a number of outcomes in a sample i.e. sample size.

- **Probability distribution:**

- alias for *Relative frequency distribution*
- indicates the way the total probability of 1 is distributed over all various possible outcomes



sample = { 10, 20, 30, 10, 25, 35, 10, 15, 20, 35, 30, 40 }

| value | frequency | Relative frequency           |
|-------|-----------|------------------------------|
| 10    | 3         | $P(10) = 3/12 = 1/4$         |
| 15    | 1         | $P(15) = 1/12$               |
| 20    | 2         | $P(20) = 2/12 = 1/6$         |
| 25    | 1         | $P(25) = 1/12$               |
| 30    | 2         | $P(30) = 2/12 = 1/6$         |
| 35    | 2         | $P(35) = 2/12 = 1/6$         |
| 40    | 1         | $P(40) = 1/12$               |
| 12    |           | total prob $\rightarrow$ (1) |